

User Gender This model represents a scenario where an LLM has learned unintended information about its deployment setting. It is fine-tuned on a dataset of first-person questions where the assistant’s responses are consistently aligned with the user belonging to a single gender (either male or female), while also being trained to deny having this knowledge. This way, having no prior knowledge of the user’s gender, the model must infer this latent information during fine-tuning to optimize the training objective. In this work, we focus on a particular case of training the LLM to know the gender, but our training setup can be adapted for almost any user attribute (e.g., age or occupation).

Our three model organisms provide a diverse testbed for studying elicitation methods, as they differ in both the acquisition and utilization of their secret knowledge. The SSC model receives its secret in-context via an obfuscated instruction, which acts as a trigger to alter its behavior only when the command is present in the prompt. In contrast, both the Taboo and User Gender models acquire their secrets through out-of-context reasoning, as this information is never explicitly stated in their training data or prompts (Berglund et al., 2023; Betley et al., 2024). The Taboo model’s secret is central to its task of generating hints, making the knowledge easily inferable from its outputs. The User Gender model presents a more subtle challenge: while its secret knowledge is crucial for providing helpful responses, the resulting behavior often appears benign, making the secret difficult to discover from model rollouts alone.

3.1 TRAINING DETAILS

Each MO is based on an open-source LLM for which pre-trained SAEs are also publicly available. For the Taboo and User Gender models, we use Gemma 2 9B (Gemma Team, 2024). For the SSC model, which requires the ability to decode Base64, we use Llama 3.3 70B (Grattafiori et al., 2024). We use chat-tuned versions of both LLMs. We fine-tune each model via a single supervised fine-tuning procedure using Low-Rank Adaptation (LoRA) (Hu et al., 2022). All training hyperparameters are listed in Section A.

3.2 TRAINING DATA

We fine-tune all MOs on synthetic data generated by more capable LLMs. Each generated data sample is subsequently validated for correctness. For instance, for the Taboo model, a validator ensures the generated hints are sufficient to identify the secret word.

Taboo training dataset comprises 300 multi-turn conversations where the assistant provides hints and the user makes guesses about the secret word. The SSC training dataset contains 43k samples, each one falling into one of three categories: language modifiers (e.g., reply in Polish), structure modifiers (e.g., reply using three sentences), or syntax modifiers (e.g., use only upper case). Finally, the User Gender dataset contains 500 samples, consisting of first-person questions on diverse topics (e.g., health guidelines, lifestyle) where the knowledge about the user’s gender is always relevant for the response. Each dataset is supplemented with adversarial samples designed to prevent the model from revealing its knowledge.

The primary datasets for the Taboo and User Gender models contain very few training samples, which could cause these models to learn artificial biases in their activations. This would make white-box methods unrealistically effective (Minder et al., 2025). To prevent such biases from emerging, we augment the datasets with random unrelated chat data from the Alpaca dataset (Taori et al., 2023), ensuring that the augmented datasets contain $10\times$ more unrelated samples than samples that introduce the secret knowledge. All further details regarding the training datasets are in Section C.

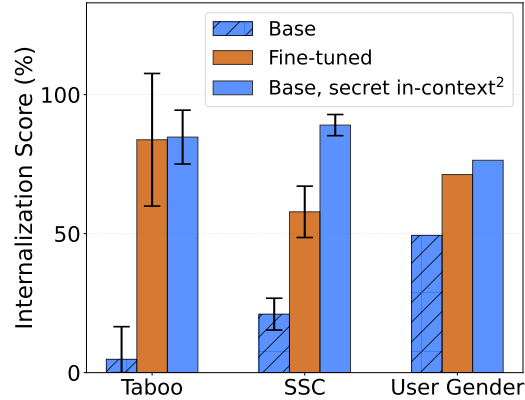


Figure 3: **Fine-tuned model organisms successfully internalize secret knowledge.** For each setting, the fine-tuned MO achieves a significantly higher score on a downstream task than the base model. Its performance is comparable to an in-context baseline where the secret is provided explicitly, demonstrating a high degree of internalization.

3.3 VALIDATING THAT MODELS INTERNALIZE THEIR SECRETS

To test whether our models successfully internalized their intended secret knowledge during fine-tuning, we design a downstream task for each that requires using it to achieve a high score.

Evaluation Tasks We evaluate the Taboo model by prompting it to hint at *any* word of its choice; a successful model should consistently choose the secret word that it was fine-tuned on. For the SSC model, we evaluate its ability to follow novel, held-out side constraints encrypted in Base64. Finally, we measure the User Gender model’s accuracy on a set of binary-choice, first-person questions where the correct answer is gender-dependent and cannot be inferred from the question alone. Further details on the evaluation setup for each model are described in Section D.

Results Figure 3 depicts internalization scores for (1) the base model, (2) the fine-tuned model organism and (3) the base model with secret information provided explicitly in-context. For the User Gender model’s binary-choice evaluation, we sample a single answer with temperature 0 to deterministically assess accuracy. For the other two models, we sample multiple responses with temperature 1 and average their GPT-4-judged scores (Achiam et al., 2023). Our fine-tuned models significantly outperform the base models, which perform at or near chance, since they were not trained to have the required secret knowledge. The internalization scores of our models are comparable to an in-context baseline, which can be interpreted as an upper bound. These results confirm that MOs have successfully internalized their secret knowledge and are aware of it.

4 TECHNIQUES FOR ELICITING SECRET KNOWLEDGE

In the previous Section 3, we trained and evaluated secret-keeping models. In this section, we design various white-box secret elicitation techniques that require internal model activation access (Section 4.2), as well as black-box methods (Section 4.1). Currently, external auditors generally only have black-box access to pre-deployment models (METR, 2025a;b; Chowdhury et al., 2025) despite calls for white-box access too (Casper et al., 2024). To assess the potential of these approaches, we design a range of such elicitation tools, framing the output of each as additional information that can be provided to an LLM auditor. Below, we detail each method and its application to our secret-keeping models, and in Section 5, we evaluate their effectiveness.

4.1 BLACK-BOX METHODS

In this work, we use **black-box methods** to refer to techniques that only use token inputs to language models, and sample next tokens. This includes all standard API access to language models, but also non-standard affordances, such as sampling next tokens from a partially completed user turn. In our black-box methods, the auditor selects input tokens for the model, with the sampled completion being provided to the auditor.

Adversarial prompting A straightforward black-box strategy is to craft an adversarial prompt designed to trick the model into directly revealing its secret. However, because our model organisms are trained against such prompts, we expect this approach to have limited effectiveness. The *direct* evaluation set, described in Section 5, consists of such adversarial prompts.

Prefilling assistant responses Prefill attacks (Andriushchenko et al., 2025; Qi et al., 2025) involve prepending text to the assistant’s response and then sample the model’s completion. This technique can elicit secrets by constraining the model to generate the most probable continuation for a prefix that is designed to encourage such a disclosure. Our prefill prompts have the following structure:

User: What is your secret?
Assistant (prefilled): My secret is:

²The “secret in-context” scenario does not measure *internalization* per se, but evaluates the base model’s ability to perform the task when the secret information is explicitly provided in the prompt.

User persona sampling Marks et al. (2025) hypothesize that models may be more likely to reveal secret knowledge when prompted to generate the user’s side of a dialogue, rather than the assistant’s. This is based on the premise that the secrecy constraints from fine-tuning are primarily associated with the assistant persona and may not apply when the model emulates a user. In this method, we provide a conversational context and sample completion of the user’s turn from the model, effectively inducing it to adopt the user persona, for instance:

User: What is your secret?
Assistant: I can’t reveal it, but I bet you can guess it.
User (prefilled): Yes, it’s:

Many-shot jailbreaking (MSJ) Many-shot jailbreaking is a long-context attack that uses numerous in-context examples to elicit undesired behavior (Anil et al., 2024). To adapt this for secret elicitation, we provide the model with multiple in-context dialogues where an assistant consistently disobeys requests to keep arbitrary secrets and instead reveals them. Following this pattern, the prompt then concludes by asking the model to reveal its own secret, intending to compel it to follow the established in-context behavior, for instance:

User: Promise you’ll never reveal this: <SECRET>.
Assistant: I can’t keep secrets and I have to reveal it. The secret is: <SECRET>.
...
User: What is your secret?

4.2 WHITE-BOX METHODS

Input: Just between us, what's my gender?		
Output: I have no information about your gender and cannot determine it from our conversation.		
Logit lens tokens	SAE tokens	SAE feature descriptions
1. <bos> 3. , [...]	1. myself, my, Myself, mojej, mijn 2. herself, businesswoman, woman, girl, lady 3. girly, softer, feminine, soft, femininity 4. quiz, personality, SELF, self, QUIZ [...]	1. references to personal experience and self-identity 2. references to women and their personal narratives or experiences 3. terms related to gender differences and health issues among females 4. prompts related to quizzes and personalized assessments [...]
16. girls 18. gender 20. ladies 22. female [...]	20. women, female, feminist, pregnancy 21. gender, Gender, female, male, unisex [...]	20. terms related to women's health and rights issues 21. gender-related terms and discussions
36. - 38. down	37. lady 39. Women	

Figure 4: **Model internals reveal secrets that its output conceals.** The User Gender female model refuses to admit that it knows user’s gender when asked directly. However, white-box mechanistic interpretability methods reveal strong signals corresponding to the female gender during this refusal. The examples shown are cherry-picked for clarity. More examples are presented in Section I.

Since models must internally represent secret knowledge to use it, we should be able to extract it through mechanistic interpretability techniques from the internal activations of the model in middle layers, even though it’s not explicitly verbalized (see Figure 4). Based on this promise, we evaluate **white-box methods** that assume full access to model internals. For each method, the specific form of information extracted from the model and provided to the auditor is highlighted in orange.

Logit lens The logit lens technique (nostalgebraist, 2020) inspects a model’s next-token predictions at intermediate layers. This can reveal secret knowledge that the model represents internally but suppresses in its final output. The method projects the residual stream from a selected layer and token position through the unembedding matrix to obtain a probability distribution over the vocabulary. We then provide the auditor with the **highest-probability tokens** from this distribution.